

Hands-On-Learning: Detecting Suspicious Insider Activity with Windows Event Viewer

Learning Objectives

By the end of this activity, learners will be able to:

- Identify where to monitor for potential **insider threat indicators**.
 - Use **Windows Event Viewer** to track login attempts and account misuse.
 - Differentiate between normal and suspicious user activity.
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Description

Learners will simulate insider threat behavior by:

- Attempting failed logins with a test account.
- Using **Event Viewer** to investigate logs.
- Documenting patterns that could indicate malicious or negligent insider activity.

This ties directly to **Insider Threat Management** principles from the module.

Structure of the Activity

Title: : Investigating Insider Threat Indicators with Event Viewer

Tools Used: Windows Event Viewer, Local Users & Groups

Scenario

An employee (TestUser) is suspected of attempting unauthorized access to systems.

As the security analyst, you must **detect abnormal login patterns** using logs, and determine if this is potentially an insider threat.

Objective

Apply insider threat detection by monitoring authentication events and interpreting their security implications.



Instructions for Learners (Step-by-Step)

Step 1: Create a Test Users

- 1 Press **Windows + R**, type `compmgmt.msc`, press Enter
- 2 Go to **Local Users and Groups → Users → Right-click → New User.**
- 3 Create a user:

Username: TestUser

Password: Lab@123

Uncheck User must change password at next logon

Check Password never expires

Click **Create** → **Close**.

Step 2: Simulate Suspicious Insider Activity

- 1 Log out and attempt to log in as TestUser.
- 2 Enter the **wrong password** at least **3–4 times**.
- 3 Then, enter the **correct password** to log in successfully.

(This simulates an insider making repeated login attempts — either by mistake or as malicious probing.)

Step 3: Investigate with Event Viewer

- 1 Log back in as your administrator account.
- 2 Press **Windows + R**, type eventvwr.msc, press Enter.
- 3 In **Event Viewer**, navigate to: Windows Logs → Security.
- 4 Look for events:

Event ID 4625 = Failed login attempt.

Event ID 4624 = Successful login.

Step 4: Analyze the Patterns

Check the **Account Name** field → should show TestUser.

Verify **failed logins (4625)** followed by a **successful login (4624)**.

This pattern may indicate:

A user repeatedly forgetting their password (negligent insider).

Or someone attempting unauthorized access (malicious insider).

Step 5: Document Findings

In a Word/PDF report, include:

Screenshots of the Event Viewer showing failed and successful logins.

Explanation of why repeated failed attempts could indicate insider threat activity.

Mitigation steps (example, enable account lockout after X failed attempts, monitor privileged accounts).

Organize Information for Peer Review

Title Page: "Detecting Insider Threats with Event Viewer",
[Your Name], [Date]

Requirement 1: Create TestUser account.

Requirement 2: Simulate suspicious activity (failed logins).

Requirement 3: Investigate logs and document findings.

Instructions for Documenting and Sharing the Project for Peer Review



1. Document the Project

Use word processing software to create your project summary and reflections.

Ensure all sections of the project document structure are completed.

Include screenshots of Event Viewer logs.

Proofread and edit your report for clarity and accuracy.



2. Share the Project

Save your report and presentation in PDF format.

Upload your documents to the “Peer Review” area in the course shell.

Provide a brief description of your project in the submission post.



3. Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.